

HealthClaimHardened v1

Thesis-quality local ZKP measurements for the legacy interval circuits, HealthClaimCore, HealthClaimHardened, controlled Merkle and policy components, and a local proving pipeline.

Decision	Scope
PASS WITH OBSERVATIONS	Reproducible local benchmarks; no Gateway, Fabric, network, Docker, IPFS, or federated-learning measurements.

Executive results

HealthClaimCore measured 19,657 constraints and 65.811 ms mean proving time; HealthClaimHardened measured 22,021 constraints and 73.575 ms. Hardened added 2,364 constraints (+12.02%) and 7 public inputs, while mean proving rose 7.764 ms (+11.80%). Verification remained about 0.7 ms, and the BN254 Groth16 proof stayed 164 bytes.

The depth 4/8/16 Merkle-only and policy set size 4/8/16/32 experiments are isolated circuit variants, not full Z7 scaling. Their constraints and proving time grew with component size, while verifier latency and proof size stayed nearly flat at fixed public-input count.

Measurement date: 2026-10-04
Repository base commit: ee4df56b09ea7f0abbe66f31a8e6e429ec401a0f (Z8); benchmark harness was an uncommitted Z9 working-tree change at measurement time and is included in the Z9 commit.
Go 1.25.8 for Core/Hardened; Go 1.22.2 for legacy.

1. Objective, environment, and method

The purpose was to measure proving and verification costs reproducibly without changing production Z6 or Z7 circuit semantics. Environment: Ubuntu 24.04.5, Linux 6.17.0-1032-oem, linux/amd64; AMD Ryzen 7 PRO 8840HS, 8 cores / 16 threads; 27 GiB RAM; GOMAXPROCS=16. Core/Hardened dependencies: gnark 0.16.3 and gnark-crypto 0.21.0. Legacy: gnark 0.9.1 and gnark-crypto v0.12.2-0.20231013160410-1f65e75b6dfb.

Core, Hardened, and legacy used 10 independent runs, 10 timed operations each, with Go -benchmem. Scaling components used 5 runs, 5 operations each. Compile/setup used 3 one-operation samples. Witness and full local pipeline used 10 runs, 10 operations each. Compile, Groth16 setup, witness preparation, and warm-up proof are outside prove/verify benchmark loops.

Means, medians, sample standard deviations, and nearest-rank P95 are over independent Go benchmark runs. At n=10 or n=5, P95 equals the maximum observation. Compile/setup have n=3, so their P95 is also the maximum. The raw output log records exact commands, versions, commit, and working-tree state at launch.

2. E1: legacy interval baseline

The legacy implementation proves only ``MinAge <= Age <= MaxAge`` and ``CategoryMin <= DiagnosisCode <= CategoryMax``. These numeric intervals are not equivalent to signed records, Merkle membership, policy commitments, or contextual claim proofs. Each interval circuit compiled to 3,754 constraints, 2 public / 1 private input, and a 164-byte proof.

Legacy circuit / stage	Mean	Median	SD	P95	B/op	allocs/op
AgeRange prove	8.781 ms	8.645 ms	0.509 ms	10.077 ms	1,840,438	1,638.5
AgeRange verify	0.678 ms	0.671 ms	0.047 ms	0.767 ms	29,000	156.4
DiagnosisCategory prove	8.638 ms	8.580 ms	0.198 ms	9.131 ms	1,839,602	1,634.1
DiagnosisCategory verify	0.687 ms	0.686 ms	0.048 ms	0.790 ms	28,816	156.0

3. E2/E3: complete Core and Hardened circuits

Metric	Core (Z6)	Hardened (Z7)	Difference	Change
Constraints	19,657	22,021	+2,364	+12.02%
Public variables	6	13	+7	+116.67%
Private variables	31	31	0	0%
Prove mean	65.811 ms	73.575 ms	+7.764 ms	+11.80%
Verify mean	0.708 ms	0.693 ms	-0.015 ms	-2.12%*
Proof size	164 bytes	164 bytes	0	0%
Prove B/op	21,151,356	21,689,935	+538,579	+2.55%
Prove allocs/op	2,602.8	2,605.2	+2.4	+0.09%
Verify B/op	30,496	32,024	+1,528	+5.01%
Verify allocs/op	165.1	180.3	+15.2	+9.21%

* The apparent verification reduction is within sub-millisecond run variability and is not evidence Hardened verifies faster. Core proof size was independently measured in the Z9 metadata/proof round-trip.

4. Compilation and Groth16 setup

These stages were timed separately from proof creation. Setup and compile measurements each have three one-operation samples.

Circuit stage	Mean	Median	SD	P95	B/op	allocs/op
Core compile	70.153 ms	69.792 ms	1.155 ms	71.445 ms	103,924,957	549,827
Hardened compile	73.421 ms	72.545 ms	2.701 ms	76.451 ms	109,236,883	595,870
Core Groth16 setup	650.110 ms	648.977 ms	13.030 ms	663.669 ms	35,111,813	580
Hardened Groth16 setup	735.778 ms	725.138 ms	27.449 ms	766.954 ms	37,380,752	523

5. E4: Merkle-depth scaling

Benchmark-only relation: Poseidon2 Merkle path hashing, Boolean direction bits, and root equality. It omits signature, record leaf, nullifier, policy, and context gadgets. Public variables = 1; private variables = 1 + 2 x depth.

Depth	Constraints	Private	Prove mean / median / SD / P95	Verify mean / median / SD / P95	Prove B/op; allocs	Verify B/op; allocs	Proof
4	1,501	9	11.865 / 11.880 / 0.076 / 11.932 ms	0.643 / 0.652 / 0.045 / 0.682 ms	1,006,020; 1,947	29,697; 155	164 B
8	3,001	17	21.438 / 21.024 / 1.000 / 23.186 ms	0.658 / 0.622 / 0.066 / 0.758 ms	1,792,361; 2,357	29,464; 155	164 B
16	6,001	33	33.790 / 32.655 / 2.725 / 38.410 ms	0.674 / 0.684 / 0.028 / 0.697 ms	3,202,291; 2,419	29,464; 155	164 B

Constraint growth is linear in depth, approximately 1,500 per four levels. Proving time rises with path depth; verification stays about flat because the public-input count remains fixed. Groth16 proof size remains 164 bytes.

6. E5: policy-size scaling

Benchmark-only relation: Poseidon2 policy commitment, diagnosis membership, and lab interval. It omits 64-bit decomposition, coverage/claim checks, record/signature/path, and context. Public variables = 1; private variables = set size + 5.

Entries	Constraints	Private	Prove mean / median / SD / P95	Verify mean / median / SD / P95	Prove B/op; allocs	Verify B/op; allocs	Proof
4	2,776	9	14.887 / 14.695 / 0.487 / 15.753 ms	0.650 / 0.637 / 0.040 / 0.719 ms	1,651,733; 2,453	29,464; 155	164 B
8	3,528	13	17.895 / 17.780 / 0.341 / 18.463 ms	0.630 / 0.634 / 0.028 / 0.664 ms	1,843,557; 2,487	29,464; 155	164 B
16	5,032	21	27.342 / 25.212 / 5.235 / 36.676 ms	0.697 / 0.701 / 0.061 / 0.786 ms	2,884,728; 2,636	29,464; 155	164 B
32	8,040	37	35.262 / 34.736 / 1.532 / 37.743 ms	0.707 / 0.720 / 0.034 / 0.744 ms	3,603,213; 2,806	29,786; 156	164 B

The 16-entry sample has a high proving outlier; the cause was not instrumented. Constraint and proving cost grow with set size, while verification remains nearly flat at fixed public-input count.

7. E6/E7: context and witness complexity

E6 uses the observed complete-circuit Core-to-Hardened comparison as an aggregate context/version overhead. It does not isolate each added context field. This aggregate adds 2,364 constraints and 7 public inputs, costs 7.764 ms more mean proving time, leaves private witness count unchanged, and leaves verification and 164-byte proof size nearly unchanged.

E7 uses the isolated Merkle and policy variants above. Record signature and context hashing remain part of both complete circuits. Signature-only cost was NOT EXECUTED because an artificial signature-only circuit would not represent its composition in the signed-record claim relation. No standalone signature-cost claim is made.

8. E8/E9: repeated-run stability and local pipeline

Operation	Mean	Median	SD	P95	B/op	allocs/op
Witness / record construction	0.339 ms	0.335 ms	0.038 ms	0.405 ms	18,402	524
Full local witness -> prove -> verify	76.320 ms	76.001 ms	1.380 ms	79.179 ms	21,740,249	3,312

The witness benchmark includes native record/policy/context construction, authority signing, Merkle root/nullifier derivation, and witness encoding. The full local pipeline adds Groth16 proving and verification. Both exclude compile/setup. There is no Gateway/Fabric/network/Docker/IPFS/federated-learning latency.

Stability and interpretation

Core proving showed a high P95 (72.338 ms vs 65.108 ms median); Hardened proving was tighter in this run (75.972 vs 73.494 ms). The policy 16-entry experiment also had a high P95 (36.676 vs 25.212 ms median). Since the runs collect no GC trace, CPU frequency, or scheduler profile, these outliers cannot be attributed to a particular cause. ``-benchmem`` reports allocation volume/count, not peak resident memory.

Reproducibility

From repository root: `GOMAXPROCS=16 ./zkp/z9bench/run.sh`. The script saves raw outputs and uses explicit GOCACHE locations in /tmp. It reports the Z8 base commit and working-tree state because the benchmark harness itself was being developed in the Z9 working tree. The Z9 module locally replaces the Core and Hardened modules; ``go mod verify`` cannot verify zip hashes for local replacements. The production Core and Hardened modules each passed ``go mod verify``.

All final test commands passed: Core, Hardened (including Z8 adversarial tests), legacy module, and Z9 benchmark harness.

PASS WITH OBSERVATIONS

This evaluation completes the requested Z9 performance phase. Component variants are labeled honestly and no distributed-system performance is claimed.